COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	08/26/2026	**DATE SUBMITTED:**	08/26/2026
GROUP	PE 6	**INSTRUCTOR:**	ENGR. JIMLORD QUEJADO

PRELIM LABORATORY PRELIM

1. PROCEDURE OUTPUT

What procedures did you perform?

Problem Title: The Customer Service Verification Problem

A customer service center processes customers in the order they arrive. Each customer is assigned a unique customer number.

At the end of the day, the supervisor wants to verify whether the order in which the customers were served is the **reverse of the order in which they originally arrived**.

Given a sequence of customer numbers, your task is to determine whether the sequence is the same when its order is reversed.

For example:

101 102 103 102 101

is a valid reverse-order sequence because the values read from the beginning and from the end are the same.

Your task is to use **two data structures** for your solution:

- Stack
- Queue

However, the two data structures **must have different internal representations**, either:

- Array-based
- Linked List

Example: If your stack is array-based, your queue must be implemented using a linked list, and vice versa.

Each customer number must be inserted into **both the Stack and Queue**.

You must then use the behavior of the two data structures to determine whether the sequence satisfies the required condition.

You may only use the **public operations provided by the existing Stack and Queue header files**. Do not modify the internal representation of either data structure.

The user is then prompted whether the given customer sequence satisfies the reverse-order condition.

Sample Test Data

Customer Numbers	Valid?	Output (Console)	Remark
101 102 103 102 101	Yes		
101 102 103	No		
205 205	Yes		
501	Yes		
10 20 30 40 50	No		
11 22 33 22 11	Yes		

Submission Requirements:

1. Activity template containing the screenshots and analysis of the function you created to solve the problem. (Code snippet only, not the whole code)

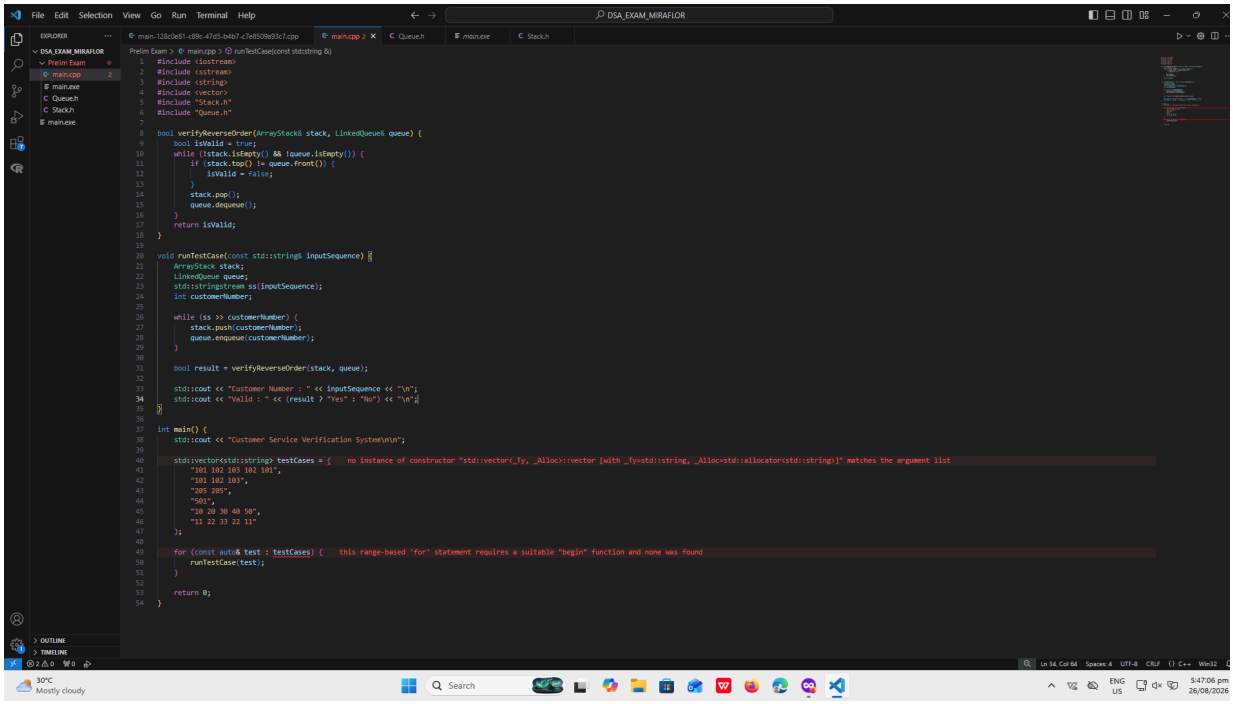
2. Code outputs inside the sample test data

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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```cpp
#include <iostream>
#include <sstream>
#include <string>
#include <vector>
#include "Stack.h"
#include "Queue.h"

bool verifyReverseOrder(ArrayStack& stack, LinkedQueue& queue) {
    bool isValid = true;
    while (!stack.isEmpty() && !queue.isEmpty()) {
        if (stack.top() != queue.front()) {
            isValid = false;
        }
        stack.pop();
        queue.dequeue();
    }
    return isValid;
}

void runTestCase(const std::string& inputSequence) {
    ArrayStack stack;
    LinkedQueue queue;
    std::stringstream ss(inputSequence);
    int customerNumber;

    while (ss >> customerNumber) {
        stack.push(customerNumber);
        queue.enqueue(customerNumber);
    }

    bool result = verifyReverseOrder(stack, queue);

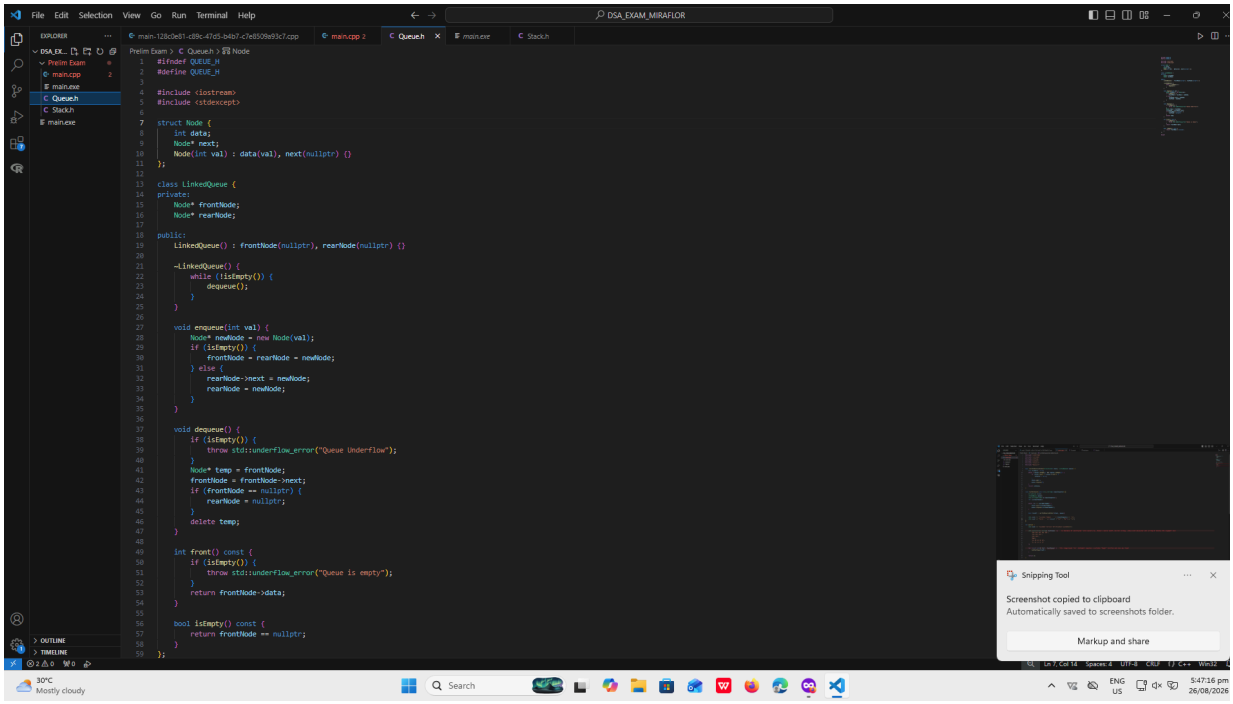
    std::cout << "Customer Number : " << inputSequence << "\n";
    std::cout << "Valid : " << (result ? "Yes" : "No") << "\n";
}

int main() {
    std::cout << "Customer Service Verification System\n\n";

    std::vector<std::string> testCases = {    no instance of constructor "std::vector<_Ty, _Alloc>::vector [with _Ty=std::string, _Alloc=std::allocator<std::string>]" matches the argument list
        "101 102 103 102 101",
        "101 102 103",
        "205 205",
        "501",
        "10 20 30 40 50",
        "11 22 33 22 11"
    };

    for (const auto& test : testCases) {    this range-based 'for' statement requires a suitable "begin" function and none was found
        runTestCase(test);
    }

    return 0;
}
```



```cpp
#ifndef QUEUE_H
#define QUEUE_H

#include <iostream>
#include <stdexcept>

struct Node {
    int data;
    Node* next;
    Node(int val) : data(val), next(nullptr) {}
};

class LinkedQueue {
private:
    Node* frontNode;
    Node* rearNode;

public:
    LinkedQueue() : frontNode(nullptr), rearNode(nullptr) {}

    ~LinkedQueue() {
        while (!isEmpty()) {
            dequeue();
        }
    }

    void enqueue(int val) {
        Node* newNode = new Node(val);
        if (isEmpty()) {
            frontNode = rearNode = newNode;
        } else {
            rearNode->next = newNode;
            rearNode = newNode;
        }
    }

    void dequeue() {
        if (isEmpty()) {
            throw std::underflow_error("Queue Underflow");
        }
        Node* temp = frontNode;
        frontNode = frontNode->next;
        if (frontNode == nullptr) {
            rearNode = nullptr;
        }
        delete temp;
    }

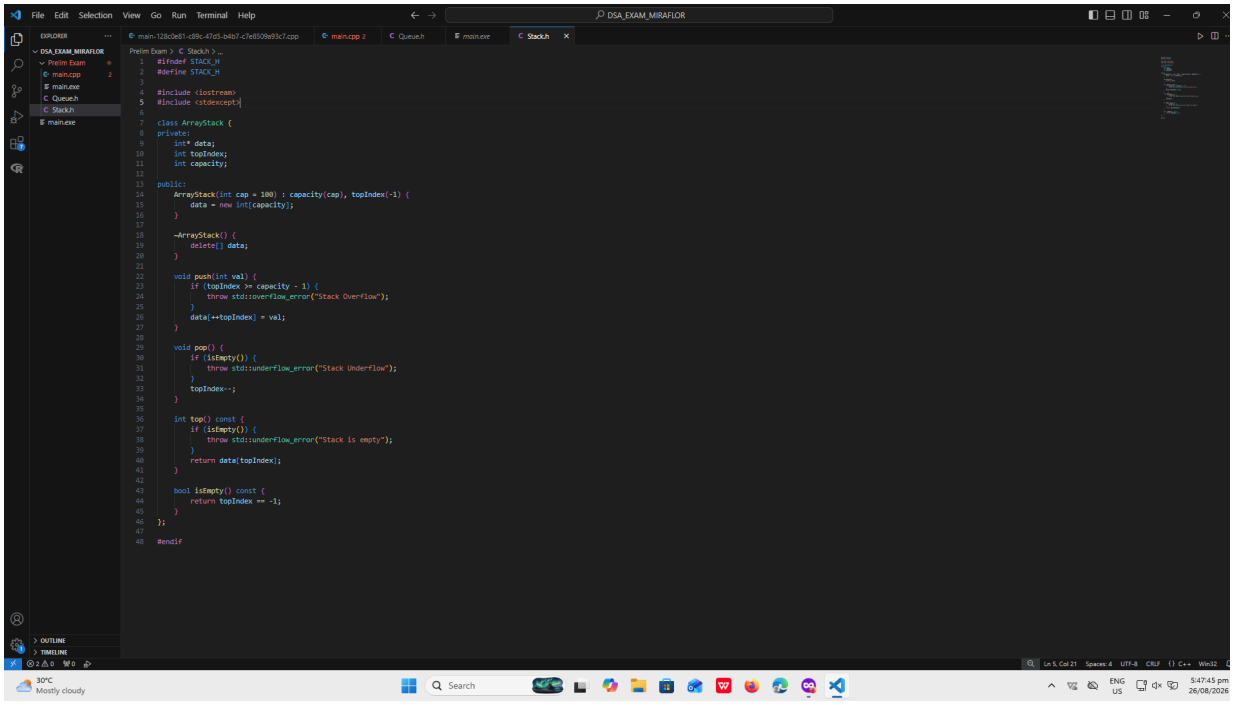
    int front() const {
        if (isEmpty()) {
            throw std::underflow_error("Queue is empty");
        }
        return frontNode->data;
    }

    bool isEmpty() const {
        return frontNode == nullptr;
    }
};
```

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```
#ifndef STACK_H
#define STACK_H

#include <iostream>
#include <stdexcept>

class ArrayStack {
private:
    int* data;
    int topIndex;
    int capacity;

public:
    ArrayStack(int cap = 100) : capacity(cap), topIndex(-1) {
        data = new int[capacity];
    }

    ~ArrayStack() {
        delete[] data;
    }

    void push(int val) {
        if (topIndex >= capacity - 1) {
            throw std::overflow_error("Stack Overflow");
        }
        data[++topIndex] = val;
    }

    void pop() {
        if (isEmpty()) {
            throw std::underflow_error("Stack Underflow");
        }
        topIndex--;
    }

    int top() const {
        if (isEmpty()) {
            throw std::underflow_error("Stack is empty");
        }
        return data[topIndex];
    }

    bool isEmpty() const {
        return topIndex == -1;
    }
};

#endif
```

```
PS C:\Users\TIPQC\Desktop\DSA_EXAM_MIRAFLOR> cd "c:\Users\TIPQC\Desktop\DSA_EXAM_MIRAFLOR\Prelim Exam\"
Customer Service Verification System

Customer Number : 101 102 103 102 101
Valid : Yes
Customer Number : 101 102 103
Valid : No
Customer Number : 205 205
Valid : Yes
Customer Number : 501
Valid : Yes
Customer Number : 10 20 30 40 50
Valid : No
Customer Number : 11 22 33 22 11
Valid : Yes
PS C:\Users\TIPQC\Desktop\DSA_EXAM_MIRAFLOR\Prelim Exam>
```

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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